

Digital Signal Analysis and Processing Numerical Companion

CT 704 (BEI/BCT, IV-I) / EX 753 (BEX, IV-II) · TU, IOE · tiers over 45 papers, 2066–2082 BS

Ch 5 • FIR Filter Design

Numerical Problems & Solutions

266 numerical marks, 7.3% of the archive, 49% of this chapter · **47 of the 70 questions are one design calculation.** The other 23 are the Remez algorithm, which asks for a flowchart and never for a number: that is \$5.5 of the theory, not here. Every design paper is below, each with its own question as printed.

The symbols, once, so nothing below is a bare number

- ω_p, ω_s pass-band and stop-band edge, radians per sample.
- $\Delta\omega = \omega_s - \omega_p$ transition width. **This, not the attenuation, sets the length.**
- A_s required stop-band attenuation, dB. **This, not the width, sets the window.**
- c the chosen window's transition constant from the table, so that its transition width is $c\pi/M$.
- M filter order; $N = M + 1$ number of taps; $\alpha = (N - 1)/2$ group delay in samples.
- ω_c cut-off of the ideal prototype; $h_d[n]$ its impulse response; $w[n]$ the window; $h[n] = h_d[n] w[n]$ the filter.

The recipe, used in §1, §2 and §3

1. **Convert the spec** to ω in radians per sample. From Hz, $\omega = 2\pi f/f_s$; from rad/s, $\omega = \Omega/f_s$; from a ripple, $A_s = -20 \log_{10} \delta_s$.
2. **Window** from A_s , taking the cheapest row of the table that clears it.
3. **Length** $M = c\pi/\Delta\omega$, rounded up; then $N = M + 1$, rounded up again to the next **odd** value.
4. **Cut-off** $\omega_c = \frac{\omega_p + \omega_s}{2}$, the middle of the transition band.
5. **Coefficients** $h[n] = h_d[n] w[n]$, with $h_d[n] = \frac{\sin(\omega_c(n - \alpha))}{\pi(n - \alpha)}$ and the centre tap $h_d[\alpha] = \omega_c/\pi$ done separately, because the general formula is 0/0 there.

The table, again, because everything turns on it

Window	Stop band	Transition $c\pi/M$
Rectangular	21 dB	$1.8\pi/M$
Bartlett	25 dB	$6.1\pi/M$
Hann	44 dB	$6.2\pi/M$
Hamming	53 dB	$6.6\pi/M$
Blackman	74 dB	$11\pi/M$

Checks that cost nothing

1. The centre tap is always ω_c/π exactly. **If it is not, ω_c is wrong.**
2. $h[n]$ must be **symmetric** about α : compute one half and mirror it.
3. A **Hann**-windowed filter has $h[0] = h[N - 1] = 0$ exactly, because the window is zero there. So "the first three coefficients" begins with a zero.
4. N odd, always, so α is an integer.

1 Fixed-window low-pass designs

TOP 13/45 79 Ba **5+3**, 71 Ch, 69 Ch **8**, 74 Ash **2+6**, 79 Ch **8**, 75 Bh **6**, 82 Ba **2+6**, 67 Mng **8**, 70 Asa **9**, 69 Bh **8**, 68 Bh **9**, 76 Ch **6**, 81 Ch **2+6**, 73 Ma **4+5**, 82 Bh **10**

QUESTION AS PRINTED 2079 Chaitra · 2075 Bhadra · [8] [6]

Find out the first three coefficients of impulse response of a low pass FIR filter having Pass band edge frequency $\omega_p = 0.24\pi$, Stop band edge frequency $\omega_s = 0.54\pi$ and Stop band attenuation $\alpha_s = 42.2$ dB using any suitable window function.

Step 1: the window, from the attenuation alone

- Required $A_s = 42.2$ dB. Read down the table to the first row that clears it: Bartlett gives 25 dB and fails, **Hann gives 44 dB and clears. So Hann**, and Hamming's 53 dB would be over-design.

$$c = 6.2 \quad (\text{Hann})$$

Step 2: the length, from the transition width alone

$$\Delta\omega = \omega_s - \omega_p = 0.54\pi - 0.24\pi = 0.30\pi$$

$$M = \frac{c\pi}{\Delta\omega} = \frac{6.2\pi}{0.30\pi} = 20.67 \Rightarrow M = 21$$

$$N = M + 1 = 22 \Rightarrow \boxed{N = 23} \text{ (next odd), } \alpha = \frac{N-1}{2} = \mathbf{11}$$

- **Two roundings, not one.** M rounds up, then N rounds up again to odd. Stopping at $N = 22$ is the commonest slip in this chapter.

Step 3: the cut-off, in the middle

$$\omega_c = \frac{\omega_p + \omega_s}{2} = \frac{0.24\pi + 0.54\pi}{2} = \mathbf{0.39\pi} = 1.2252 \text{ rad/sample}$$

$$h_d[n] = \frac{\sin(\omega_c(n - \alpha))}{\pi(n - \alpha)} = \frac{\sin(0.39\pi(n - 11))}{\pi(n - 11)}, \quad h_d[11] = \frac{\omega_c}{\pi} = \mathbf{0.39}$$

Step 4: the first three coefficients

- Hann window, general form then this filter's:

$$w[n] = 0.5 - 0.5 \cos \frac{2\pi n}{N-1} = 0.5 - 0.5 \cos \frac{2\pi n}{22}$$

$$w[0] = 0, \quad w[1] = 0.020254, \quad w[2] = 0.079373$$

$$h_d[0] = 0.022865, \quad h_d[1] = -0.009836, \quad h_d[2] = -0.035350$$

$$h[n] = h_d[n] w[n] \Rightarrow \boxed{h[0] = 0, \quad h[1] = -0.000199, \quad h[2] = -0.002806}$$

- $h[0] = 0$ is not a mistake: the Hann window is exactly zero at both ends, so the first and last taps of the filter vanish. Say so, because it looks like an error otherwise.
- The rest follows by symmetry, $h[22 - n] = h[n]$, with the centre tap $h[11] = 0.39$.

Every other fixed-window design, at a glance

The index only. Each paper's own question, and the step where it differs, is worked underneath in §1.1 to §1.5.

Papers	Spec	A_s	Window	N	ω_c
79 Ba	$\omega_p = 0.2\pi, \omega_s = 0.45\pi$	51 dB	Hamming	29	0.325π
71 Ch, 69 Ch	$\omega_p = 0.3\pi, \omega_s = 0.5\pi$	40 dB	Hann	33	0.4π
74 Ash	$\omega_p = 0.2\pi, \omega_s = 0.5\pi$	41 dB	Hann	23	0.35π
79 Ch, 75 Bh	$\omega_p = 0.24\pi, \omega_s = 0.54\pi$	42.2 dB	Hann	23	0.39π
67 Mng	$\omega_p = 0.35\pi, \omega_s = 0.45\pi$	54 dB	Blackman	111	0.4π
76 Ch, 81 Ch, 73 Ma	$\delta_p = 0.101, \delta_s = 0.01,$ $\omega_p = 0.2\pi, \omega_s = 0.4\pi$	40 dB	Hann	33	0.3π
82 Bh	2 kHz, 5 kHz at $f_s = 20$ kHz ($\omega_p = 0.2\pi, \omega_s = 0.5\pi$)	42 dB	Hann	23	0.35π
82 Ba	−3 dB at $30\pi, 50\pi$ at 45π rad/s, $f_s = 100$ Hz ($0.3\pi, 0.45\pi$)	50 dB	Hamming	45	0.375π
70 Asa	Hann named, $0.25\pi, 0.35\pi,$ main lobe $8\pi/M$	—	Hann (given)	81	0.3π
69 Bh	Hann named, $0.25\pi, 0.3\pi$	—	Hann (given)	125	0.275π
68 Bh	Blackman named, $0.24\pi, 0.34\pi,$ main lobe $12\pi/M$	—	Blackman (given)	121	0.29π

1.1 The four that print the spec exactly as the worked example does

79 Ba **5+3**, 71 Ch **8**, 69 Ch **8**, 74 Ash **2+6** · same three steps, different numbers

QUESTION AS PRINTED 2079 Baishakh · [5+3]

Design a low pass digital FIR filter having Pass band edge frequency $\omega_p = 0.2\pi$, Stop band edge frequency $\omega_s = 0.45\pi$ and Stop band attenuation $\alpha_s = 51$ dB using any appropriate window function.

- **Window.** $A_s = 51$ dB. Hann's 44 dB fails, Hamming's 53 dB clears, so **Hamming** and $c = 6.6$. **This is the one place 79 Ba differs from the worked example:** the higher attenuation buys a different window.

$$\Delta\omega = \omega_s - \omega_p = 0.45\pi - 0.2\pi = 0.25\pi$$

$$M = \frac{c\pi}{\Delta\omega} = \frac{6.6\pi}{0.25\pi} = 26.4 \Rightarrow M = 27, \quad N = M + 1 = 28 \Rightarrow \boxed{N = 29}, \quad \alpha = 14$$

$$\omega_c = \frac{0.2\pi + 0.45\pi}{2} = \mathbf{0.325\pi}$$

- Hamming window is $w[n] = 0.54 - 0.46 \cos \frac{2\pi n}{N-1}$, so unlike Hann it is **0.08 at the ends, not zero**, and $h[0] \neq 0$:

$$h[0] = 0.001797, \quad h[1] = 0.001456, \quad h[2] = -0.001029$$

- The rest is §1 Step 4 unchanged, mirrored about $\alpha = 14$, centre tap $h[14] = \omega_c/\pi = 0.325$.

QUESTION AS PRINTED 2071 Chaitra · 2069 Chaitra · [8] [8]

Design a low pass FIR filter having Pass band edge frequency $\omega_p = 0.3\pi$, Stop band edge frequency $\omega_s = 0.5\pi$ and Stop band attenuation $\alpha_s = 40$ dB using any appropriate window function.

69 Ch prints the same filter as "Design a digital FIR filter for the design of the low pass filter having $\omega_p = 0.3\pi$, $\omega_s = 0.5\pi$, $\alpha_s = 40$ dB using suitable window function."

- **Window.** $A_s = 40$ dB, so **Hann** (44 dB) and $c = 6.2$.

$$\Delta\omega = 0.5\pi - 0.3\pi = 0.20\pi, \quad M = \frac{6.2\pi}{0.20\pi} = 31 \Rightarrow M = 31$$

$$N = M + 1 = 32 \Rightarrow \boxed{N = 33} \text{ (next odd)}, \quad \alpha = 16$$

- $M = 31$ **comes out exact here**, which is where the second rounding is most easily forgotten: 32 is even, so $N = 33$.

$$\omega_c = \frac{0.3\pi + 0.5\pi}{2} = \mathbf{0.4\pi}, \quad h[16] = 0.4$$

- The rest is §1 Step 4 with $\alpha = 16$ and $N - 1 = 32$ in the window.

QUESTION AS PRINTED 2074 Ashwin · [2+6]

Why Kaiser window is better than other fixed windows in FIR filter design? Find out first six coefficients of impulse response of a low pass FIR filter having Pass band edge frequency $\omega_p = 0.2\pi$, Stop band edge frequency $\omega_s = 0.5\pi$ and Stop band attenuation $\alpha_s = 41$ dB using any appropriate window function.

- **Window.** $A_s = 41$ dB, so **Hann** (44 dB) and $c = 6.2$.

$$\Delta\omega = 0.5\pi - 0.2\pi = 0.30\pi, \quad M = \frac{6.2\pi}{0.30\pi} = 20.67 \Rightarrow M = 21, \quad \boxed{N = 23}, \quad \alpha = 11$$

$$\omega_c = \frac{0.2\pi + 0.5\pi}{2} = \mathbf{0.35\pi}, \quad h[11] = 0.35$$

- **It asks for SIX coefficients, not three.** That is the only difference, and it is worth checking you did not stop at three:

n	0	1	2	3	4	5
$h_d[n]$	-0.013137	-0.031831	-0.016057	0.023387	0.044913	0.016394
Hann $w[n]$	0	0.020254	0.079373	0.172570	0.292292	0.428843
$h[n]$	0	-0.000645	-0.001274	0.004036	0.013128	0.007030

The "why Kaiser is better" two marks: a fixed window ties attenuation and transition width to one another through c , so buying attenuation lengthens the filter. **Kaiser splits them** — β buys attenuation, N buys transition width — so it gives the shortest filter that meets both.

1.2 The three whose spec is a ripple, not a decibel figure

76 Ch **6**, 81 Ch **2+6**, 73 Ma **4+5** · **one conversion, then §1 unchanged**

QUESTION AS PRINTED 2076 Chaitra · 2081 Chaitra · 2073 Magh · [6] [2+6] [4+5]

Design the FIR filter using suitable window for the specifications:

$$0.899 \leq |H(e^{j\omega})| \leq 1 \quad \text{for } |\omega| \leq 0.2\pi$$

$$|H(e^{j\omega})| \leq 0.01 \quad \text{for } 0.4\pi \leq \omega \leq \pi$$

81 Ch prefixes "Define Gibbs phenomena in FIR filter design"; 73 Ma prefixes "Explain a Remez exchange algorithm for FIR filter design".

- **The step that is missed: the spec never says "dB".** Read the two ripples off the inequalities first, then convert. The pass-band line is $1 - \delta_p$ to 1, not $1 \pm \delta_p$, so

$$\delta_p = 1 - 0.899 = \mathbf{0.101}, \quad \delta_s = \mathbf{0.01}$$

$$A_s = -20 \log_{10} \delta_s = -20 \log_{10}(0.01) = -20(-2) = \mathbf{40 \text{ dB}}$$

- **Use δ_s , never δ_p , for the window.** The window controls the stop band; $\delta_p = 0.101$ is a red herring here and would give only 19.9 dB.
- **Now the recipe is §1 verbatim.** $A_s = 40 \text{ dB} \Rightarrow$ **Hann**, $c = 6.2$:

$$\Delta\omega = 0.4\pi - 0.2\pi = 0.20\pi, \quad M = \frac{6.2\pi}{0.20\pi} = 31, \quad N = 32 \Rightarrow \boxed{N = 33}$$

$$\omega_c = \frac{0.2\pi + 0.4\pi}{2} = \mathbf{0.3\pi}, \quad \alpha = 16, \quad h[16] = 0.3$$

- The coefficients are §1 Step 4 with these three numbers substituted.

1.3 The two whose spec is in Hz and in rad/s

82 Bh 10, 82 Ba 2+6 · two different conversions, and they are not the same one

QUESTION AS PRINTED 2082 Bhadra · [10]

Design a linear phase FIR system to meet the following specifications:

Passband edge = 2 kHz Stopband edge = 5 kHz

Stopband attenuation = 42 dB Sampling frequency = 20 kHz

- **Hz to radians per sample is $\omega = 2\pi f/f_s$.** Do it before anything else; every table constant is in radians per sample.

$$\omega_p = \frac{2\pi f_p}{f_s} = \frac{2\pi(2000)}{20000} = \frac{2\pi}{10} = \mathbf{0.2\pi}$$

$$\omega_s = \frac{2\pi f_s^{(\text{edge})}}{f_s} = \frac{2\pi(5000)}{20000} = \frac{2\pi}{4} = \mathbf{0.5\pi}$$

- **From here it is §1.** $A_s = 42 \text{ dB} \Rightarrow$ **Hann** (44 dB clears, Bartlett's 25 does not), $c = 6.2$:

$$\Delta\omega = 0.5\pi - 0.2\pi = 0.30\pi, \quad M = \frac{6.2\pi}{0.30\pi} = 20.67 \Rightarrow M = 21, \quad \boxed{N = 23}, \quad \alpha = 11$$

$$\omega_c = \frac{0.2\pi + 0.5\pi}{2} = \mathbf{0.35\pi}, \quad h[11] = 0.35$$

- **Do not convert ω_c back to Hz** unless the paper asks. It wants a filter, and $h[n]$ is indexed by sample.

QUESTION AS PRINTED 2082 Baishakh · [2+6]

Differentiate between analog and digital filter. Design an FIR lowpass digital filter that will have a -3dB cut-off at $30\pi \text{ rad/s}$ and an attenuation of 50 dB at $45\pi \text{ rad/s}$. The filter is required to have a linear phase and the system uses a sampling rate of 100 Hz.

- **rad/s to radians per sample is $\omega = \Omega/f_s$, with no 2π .** The 2π is already inside Ω . Using $2\pi\Omega/f_s$ here is the classic double-counting error and gives $\omega_s > \pi$, which is impossible.

$$\omega_p = \frac{\Omega_p}{f_s} = \frac{30\pi}{100} = \mathbf{0.3\pi}, \quad \omega_s = \frac{\Omega_s}{f_s} = \frac{45\pi}{100} = \mathbf{0.45\pi}$$

- **The second missable step: " -3 dB cut-off " is the PASS-BAND EDGE here,** not ω_c . Only the 50 dB figure is an attenuation.
- **From here it is §1.** $A_s = 50 \text{ dB}$: Hann's 44 dB fails, Hamming's 53 dB clears, so **Hamming** and $c = 6.6$:

$$\Delta\omega = 0.45\pi - 0.3\pi = 0.15\pi, \quad M = \frac{6.6\pi}{0.15\pi} = 44 \Rightarrow M = 44, \quad N = M + 1 = \boxed{45}, \quad \alpha = 22$$

$$\omega_c = \frac{0.3\pi + 0.45\pi}{2} = \mathbf{0.375\pi}, \quad h[22] = 0.375$$

- Here $N = 45$ is *already* odd, so the second rounding does nothing. Check it anyway; it is the same check that saves 71 Ch.

1.4 67 Mng, the marginal case, and the marginality is the question

67 Mng 8 · one decibel nearly doubles the filter

QUESTION AS PRINTED 2067 Mangsir · [8]

Use the window method to design a digital low-pass FIR filter with Pass band frequency (ω_p) = 0.35π , Stop band frequency (ω_s) = 0.45π with stop-band attenuation of **at least** 54dB.

- **The step that is missed: "at least 54 dB" is a hard floor, and Hamming is 53.** Reading the table loosely and taking Hamming costs the whole question, and the paper chose 54 precisely to catch that.

Hamming: 53 dB < 54 dB required $\Rightarrow$ **not allowed**

Blackman: 74 dB $\geq$ 54 dB $\Rightarrow$ **Blackman**, $c = 11$

$$\Delta\omega = 0.45\pi - 0.35\pi = 0.10\pi$$

- **Work both, because the comparison is the marks.**

$$M_{\text{Hamming}} = \frac{6.6\pi}{0.10\pi} = 66 \Rightarrow N = 67 \quad (\text{if 53 dB had been enough})$$

$$M_{\text{Blackman}} = \frac{11\pi}{0.10\pi} = 110 \Rightarrow \boxed{N = 111}, \quad \alpha = 55$$

$$\omega_c = \frac{0.35\pi + 0.45\pi}{2} = \mathbf{0.4\pi}, \quad h[55] = 0.4$$

- **One decibel of specification nearly doubles the filter**, 67 taps to 111. Say that in a line: it is the whole point of the window table, and it is worth more than the arithmetic.
- The coefficients follow §1 Step 4, with the Blackman window $w[n] = 0.42 - 0.5 \cos \frac{2\pi n}{110} + 0.08 \cos \frac{4\pi n}{110}$.

1.5 The three that name the window, and two of them redefine c

70 Asa 9, 68 Bh 9, 69 Bh 8 · **no attenuation is given at all**, so there is nothing to look up

QUESTION AS PRINTED 2070 Ashad · [9]

Use Hanning window method to design a digital low-pass FIR filter with pass-band edge frequency (ω_p) = 0.25π , stop-band edge frequency (ω_s) = 0.35π where **main lobe width of Hanning window is $8\pi/M$** , M is the filter length.

- **The step that is missed: the paper redefines c .** It hands you $8\pi/M$ as the width to use. That is the *main lobe* width, not the $6.2\pi/M$ *transition* width in the table, and the marks follow the paper.

$$\Delta\omega = 0.35\pi - 0.25\pi = 0.10\pi$$

$$M = \frac{c\pi}{\Delta\omega} = \frac{8\pi}{0.10\pi} = 80 \Rightarrow \boxed{N = 81}, \quad \alpha = 40$$

$$\omega_c = \frac{0.25\pi + 0.35\pi}{2} = \mathbf{0.3\pi}, \quad h[40] = 0.3$$

- Add one line: the table's exact $c = 6.2$ would have given $M = 6.2\pi/0.10\pi = 62$ and $N = 63$. **Both conventions are in the textbooks**; naming which one you used collects the mark either way.
- $h[n] = h_d[n] w[n]$ with $w[n] = 0.5 - 0.5 \cos \frac{2\pi n}{80}$, as §1 Step 4.

QUESTION AS PRINTED 2068 Bhadra · [9]

Use Blackman window method to design a digital low-pass FIR filter with passband edge frequency $\omega_p = 0.24\pi$, stopband edge frequency $\omega_s = 0.34\pi$ where **main lobe width of Blackman window is $12\pi/M$** , M is filter length.

- Same redefinition, different constant: $c = 12$, **not the table's 11**.

$$\Delta\omega = 0.34\pi - 0.24\pi = 0.10\pi, \quad M = \frac{12\pi}{0.10\pi} = 120 \Rightarrow \boxed{N = 121}, \quad \alpha = 60$$

$$\omega_c = \frac{0.24\pi + 0.34\pi}{2} = \mathbf{0.29\pi}, \quad h[60] = 0.29$$

- The exact $c = 11$ would give $M = 110$ and $N = 111$. Note it in one line.

QUESTION AS PRINTED 2069 Bhadra · [8]

Use the Hanning window to design a digital low-pass FIR filter with Pass band frequency (ω_p) = 0.25π and Stop band frequency (ω_s) = 0.3π .

- **This one gives NO constant**, so unlike 70 Asa and 68 Bh you fall back on the table: Hann, $c = 6.2$.

$$\Delta\omega = 0.3\pi - 0.25\pi = 0.05\pi, \quad M = \frac{6.2\pi}{0.05\pi} = 124 \Rightarrow \boxed{N = 125}, \quad \alpha = 62$$

$$\omega_c = \frac{0.25\pi + 0.3\pi}{2} = \mathbf{0.275\pi}, \quad h[62] = 0.275$$

- **A 0.05π transition is why this filter is 125 taps long** on a window that was only asked for 44 dB. Halving the transition width doubles the length; the attenuation never entered the length calculation at all.

2 Short designs with the window named

PIN 3/45 80 Bh 10, 73 Bh 2+7, 76 Bh 1+2+7 · **no window choice and no length choice**: both are given

QUESTION AS PRINTED 2080 Bhadra · [10]

Design the symmetric FIR Low Pass Filter (LPF) for which the desired frequency response is expressed as $H_d(W) = e^{-jW\tau}$, $|W| \leq W_c$ and 0 elsewhere. The length of the filter should be 7 and $W_c = 1$ rad/sample. Make use of the **Hanning** window.

- **Steps 1 to 3 of the recipe are all skipped**: the window is named, N is given, and ω_c is given. Only Step 4 is left, so there is nothing to look up and nothing to round.

$$N = 7 \Rightarrow \alpha = \tau = \frac{N-1}{2} = \mathbf{3}$$

$$h_d[n] = \frac{\sin(\omega_c(n-\alpha))}{\pi(n-\alpha)} = \frac{\sin(1 \cdot (n-3))}{\pi(n-3)}, \quad h_d[3] = \frac{\omega_c}{\pi} = \frac{1}{\pi} = 0.31831$$

$$w_{\text{Hann}}[n] = 0.5 - 0.5 \cos \frac{2\pi n}{N-1} = 0.5 - 0.5 \cos \frac{2\pi n}{6}$$

$$w_{\text{Blackman}}[n] = 0.42 - 0.5 \cos \frac{2\pi n}{6} + 0.08 \cos \frac{4\pi n}{6}$$

- **τ is not a new unknown**. A symmetric linear-phase filter has group delay $\alpha = (N-1)/2$, so $\tau = \alpha = 3$. Papers that print τ are testing exactly that.

n	0	1	2	3	4	5	6
$h_d[n]$	0.01497	0.14472	0.26785	0.31831	0.26785	0.14472	0.01497
Hann $w[n]$	0	0.25	0.75	1	0.75	0.25	0
$h[n]$	0	0.03618	0.20089	0.31831	0.20089	0.03618	0
Blackman $w[n]$	0	0.13	0.63	1	0.63	0.13	0
$h[n]$	0	0.01881	0.16874	0.31831	0.16874	0.01881	0

QUESTION AS PRINTED 2073 Bhadra · [2+7]

List out the key points of windowing and design the symmetric FIR low pass filter for which desired frequency response is expressed as $H_d(\omega) = e^{-j\omega\tau}$ for $|\omega| \leq \omega_c$; 0 elsewhere. The length of the filter should be 7 and $\omega_c = 1$ radians/sample. Use **Hanning** window as a prototype.

QUESTION AS PRINTED 2076 Bhadra · [1+2+7]

What is windowing? List out the key points of windowing and design the symmetric FIR low pass filter for which desired frequency response is expressed as $H_d(\omega) = e^{-j\omega\tau}$ for $|\omega| \leq \omega_c$; 0 elsewhere. The length of the filter should be 7 and $\omega_c = 1$ radians/sample. Use **Blackman** window as a prototype.

- **73 Bh** is 80 Bh word for word with Hann, so rows 2 and 3 of the table are its answer. **76 Bh** is the same filter with **Blackman**, so rows 4 and 5 are its answer. **The $h_d[n]$ row is shared by all three**: the ideal response does not know which window follows.
- Both windows kill the end taps, so a "7-tap" filter really has **5 non-zero coefficients**. **Worth one line**: it is why Hann and Blackman waste two taps on short filters, and why the rectangular window is not always wrong.
- Every row is symmetric about $n = 3$, confirming linear phase with a group delay of exactly 3 samples.

The "what is windowing / key points" marks, worth 1 to 3 in all three papers. Answer in four: **truncation is unavoidable** because h_d is infinite; truncating abruptly causes **Gibbs ripple**; a tapered window trades **ripple for transition width**; and the window is chosen by **attenuation**, the length by **transition width**.

3 Kaiser window designs

TOP 13/45 81 Ba **10**, 81 Bh **8**, 80 Bh **2+2+2**, 80 Ba **2+8**, 79 Bh **2+8**, 78 Bh **8**, 74 Ch **9+3**, 75 Ash **8+4**, 72 Ka **6+8**, 76 Ash **2+4+4**, 73 Ch **8**, 78 Ch **8**, 66 Ma

The Kaiser formulas, symbols first

$$\delta = \min(\delta_p, \delta_s), \quad A = -20 \log_{10} \delta$$

$$\beta = \begin{cases} 0, & A \leq 21 \\ 0.5842(A - 21)^{0.4} + 0.07886(A - 21), & 21 < A \leq 50 \\ 0.1102(A - 8.7), & A > 50 \end{cases}$$

$$M = \frac{A - 8}{2.285 \Delta\omega}, \quad N = M + 1 \text{ rounded up odd}, \quad \omega_c = \frac{\omega_p + \omega_s}{2}$$

$$w[n] = \frac{I_0\left(\beta \sqrt{1 - \left(\frac{2n}{M} - 1\right)^2}\right)}{I_0(\beta)}, \quad h[n] = h_d[n] w[n]$$

- **$\Delta\omega$ here is in radians per sample, not in units of π** . The 2.285 is a dimensional constant; feeding it 0.05 instead of 0.05 π is the most common wrong answer in this section.

QUESTION AS PRINTED 2081 Baishakh · [10]

Design a linear phase FIR filter using suitable window to meet following specifications:

$$0.99 \leq |H(e^{j\omega})| \leq 1.01, \quad \text{for } 0 \leq |\omega| \leq 0.3\pi$$

$$|H(e^{j\omega})| \leq 0.01, \quad \text{for } 0.35\pi \leq |\omega| \leq \pi$$

Step 1: read the ripples off the inequalities

- Pass band is $1 \pm \delta_p$, so $\delta_p = 1.01 - 1 = 0.01$. Stop band is $\leq \delta_s$, so $\delta_s = 0.01$.

$$\delta = \min(\delta_p, \delta_s) = \min(0.01, 0.01) = \mathbf{0.01}$$

$$A = -20 \log_{10} \delta = -20 \log_{10}(0.01) = -20(-2) = \mathbf{40 \text{ dB}}$$

Step 2: the shape parameter

- $21 < A \leq 50$, so the middle branch applies:

$$\beta = 0.5842(A - 21)^{0.4} + 0.07886(A - 21) = 0.5842(40 - 21)^{0.4} + 0.07886(40 - 21)$$

$$\beta = 0.5842(19)^{0.4} + 0.07886(19) = 0.5842(3.2474) + 1.4983 = \mathbf{3.3953}$$

Step 3: the length, the cut-off and the window

$$\Delta\omega = \omega_s - \omega_p = 0.35\pi - 0.3\pi = 0.05\pi = 0.15708 \text{ rad/sample}$$

$$M = \frac{A - 8}{2.285 \Delta\omega} = \frac{40 - 8}{2.285(0.15708)} = \frac{32}{0.35893} = 89.15 \Rightarrow M = 90$$

$$N = M + 1 = \boxed{91}, \quad \alpha = \frac{N - 1}{2} = 45$$

$$\omega_c = \frac{0.3\pi + 0.35\pi}{2} = \mathbf{0.325\pi}, \quad h[45] = \frac{\omega_c}{\pi} = 0.325$$

$$w[n] = \frac{I_0\left(3.3953\sqrt{1 - \left(\frac{2n}{90} - 1\right)^2}\right)}{I_0(3.3953)}, \quad h[n] = h_d[n] w[n]$$

- β and N came from different specifications, which is the whole advantage over a fixed window: the attenuation did not force a longer filter.

Every other Kaiser paper, at a glance

Eight of the thirteen share $\delta = 0.01$, so they share $A = 40$ dB and $\beta = 3.3953$ exactly. Only $\Delta\omega$, and therefore N , changes. Each paper's own question and its differing step are in §3.1 to §3.5.

Papers	Spec	δ	A	β	N
81 Ba, 81 Bh	$0.3\pi / 0.35\pi, \delta = 0.01$	0.01	40	3.3953	91
80 Bh	$0.19\pi / 0.21\pi, \delta_p = 0.05$	0.01	40	3.3953	225
80 Ba	$0.016\pi / 0.08\pi$	0.01	40	3.3953	71
79 Bh	$0.2\pi / 0.4\pi, \delta_p = 0.101$	0.01	40	3.3953	25
78 Bh, 74 Ch	band pass, $0.25\pi / 0.35\pi$ and $0.6\pi / 0.65\pi$	0.01	40	3.3953	91
75 Ash, 72 Ka	$0.19\pi / 0.21\pi$	0.01	40	3.3953	225
76 Ash	$0.16\pi / 0.18\pi$	0.01	40	3.3953	225
73 Ch	$0.09\pi / 0.14\pi, \delta_p = 0.02$	0.01	40	3.3953	91
78 Ch	$0.19\pi / 0.21\pi, \text{both } \delta = 0.02$	0.02	33.98	2.6523	183
66 Ma	$0.25\pi / 0.65\pi, \text{both } \delta = 0.035$	0.035	29.12	1.9903	9

3.1 The five that are 81 Ba with different edges

81 Bh **8**, 80 Ba **2+8**, 79 Bh **2+8**, 75 Ash **8+4**, 72 Ka **6+8**, 76 Ash **2+4+4** · $\delta = 0.01, A = 40, \beta = 3.3953$ throughout; **only $\Delta\omega$ moves**

QUESTION AS PRINTED 2081 Bhadra · [8]

Design a low pass FIR filter using suitable window to meet following specifications:

$$0.99 \leq |H(e^{j\omega})| \leq 1.01 \quad \text{for } 0 \leq |\omega| \leq 0.3\pi$$

$$|H(e^{j\omega})| \leq 0.01 \quad \text{for } 0.35\pi \leq |\omega| \leq \pi$$

- **Identical to 81 Ba in every figure**, and it does not even say "Kaiser": you pick it, because a fixed window at $\Delta\omega = 0.05\pi$ would need $M = 6.2\pi / 0.05\pi = 124$ against Kaiser's 90.
- Answer as §3 above: $\delta = 0.01, A = 40$ dB, $\beta = 3.3953, \boxed{N = 91}, \omega_c = 0.325\pi$.

QUESTION AS PRINTED 2080 Baishakh · [2+8]

In which case do we choose FIR filter and IIR filter? Design a linear phase FIR filter using Kaiser Window to meet the following specifications:

$$0.99 \leq |H(e^{j\omega})| \leq 1.01 \quad \text{for } 0 \leq \omega \leq 0.016\pi$$

$$|H(e^{j\omega})| \leq 0.01 \quad \text{for } 0.08\pi \leq \omega \leq 2\pi$$

- **Two misprints, and both must be read through.** " $0 \leq \omega \leq 0.016\pi$ " is impossible as written; it means $0 \leq \omega \leq 0.016\pi$. " $\leq 2\pi$ " should be $\leq \pi$, since a real filter's spectrum is only defined to π . Note both in one line and design the sensible filter.

$$\delta_p = 0.01, \delta_s = 0.01 \Rightarrow \delta = 0.01, \quad A = -20 \log_{10}(0.01) = 40 \text{ dB}, \quad \beta = 3.3953$$

$$\Delta\omega = 0.08\pi - 0.016\pi = 0.064\pi = 0.20106$$

$$M = \frac{A - 8}{2.285 \Delta\omega} = \frac{32}{2.285(0.20106)} = 69.65 \Rightarrow M = 70, \quad \boxed{N = 71}, \quad \alpha = 35$$

$$\omega_c = \frac{0.016\pi + 0.08\pi}{2} = \mathbf{0.048\pi}, \quad h[35] = 0.048$$

QUESTION AS PRINTED 2079 Bhadra · [2+8]

Define Gibb's phenomenon. Design the FIR filter using Kaiser window technique for the specifications:

$$0.899 \leq |H(e^{jw})| \leq 1 \quad \text{for } |w| \leq 0.2\pi$$

$$|H(e^{jw})| \leq 0.01 \quad \text{for } 0.4\pi \leq w \leq \pi$$

- **The pass band is $1 - \delta_p$ to 1, not $1 \pm \delta_p$** , so $\delta_p = 1 - 0.899 = 0.101$. It is ten times δ_s and the min throws it away:

$$\delta = \min(0.101, 0.01) = \mathbf{0.01}, \quad A = 40 \text{ dB}, \quad \beta = 3.3953$$

$$\Delta\omega = 0.4\pi - 0.2\pi = 0.2\pi = 0.62832$$

$$M = \frac{32}{2.285(0.62832)} = 22.29 \Rightarrow M = 23, \quad N = 24 \Rightarrow \boxed{N = 25}, \quad \alpha = 12$$

$$\omega_c = \frac{0.2\pi + 0.4\pi}{2} = \mathbf{0.3\pi}, \quad h[12] = 0.3$$

- **$N = 25$ against 81 Ba's 91** on the same attenuation. A four-times wider transition band bought a filter a quarter as long; that contrast is the answer to "why Kaiser".

QUESTION AS PRINTED 2075 Ashwin · 2072 Kartik · [8+4] [6+8]

Design a linear phase FIR filter using Kaiser Window to meet the following specifications:

$$0.99 \leq |H(e^{jw})| \leq 1.01, \quad \text{for } 0 \leq w \leq 0.19\pi$$

$$|H(e^{jw})| \leq 0.01, \quad \text{for } 0.21\pi \leq w \leq \pi$$

72 Ka prints the first line as " $0 \geq w \geq 0.19\pi$ ", a misprint for $0 \leq w \leq 0.19\pi$. 75 Ash adds "Draw the flow chart for Optimum filter design"; 72 Ka opens with the same flowchart.

$$\delta = \min(0.01, 0.01) = 0.01, \quad A = 40 \text{ dB}, \quad \beta = 3.3953$$

$$\Delta\omega = 0.21\pi - 0.19\pi = 0.02\pi = 0.062832$$

$$M = \frac{32}{2.285(0.062832)} = 222.89 \Rightarrow M = 223, \quad N = 224 \Rightarrow \boxed{N = 225}, \quad \alpha = 112$$

$$\omega_c = \frac{0.19\pi + 0.21\pi}{2} = \mathbf{0.2\pi}, \quad h[112] = 0.2$$

- **225 taps is the right answer, not an arithmetic slip.** A 0.02π transition is extremely narrow; expect a three-figure N and do not "correct" it.

QUESTION AS PRINTED 2076 Ashwin · [2+4+4]

In which case do we choose FIR filter and IIR filter? Design a Kaiser Window to meet the following specifications.

$$0.99 \leq |H(e^{jw})| \leq 1.01, \quad \text{for } 0 \leq w \leq 0.16\pi$$

$$|H(e^{jw})| \leq 0.01, \quad \text{for } 0.18\pi \leq w \leq 2\pi$$

Also "Draw the flow chart for Remez-Exchange algorithm".

- Same $\Delta\omega = 0.18\pi - 0.16\pi = 0.02\pi$ as 75 Ash, so **$N = 225$ again**; only ω_c moves.

$$\omega_c = \frac{0.16\pi + 0.18\pi}{2} = \mathbf{0.17\pi}, \quad \alpha = 112, \quad h[112] = 0.17$$

- The " $\leq 2\pi$ " is the same misprint as 80 Ba; read it as π .

3.2 80 Bh asks for three numbers and no filter

80 Bh **2+2+2** · **and its two ripples are different**, which is where the min finally matters

QUESTION AS PRINTED 2080 Bhadra · [2+2+2]

Kaiser window is to be used to design a linear phase FIR filter that meets following specification:

$$|H(e^{j\omega})| \leq 0.01, \quad 0.21\pi \leq |\omega| \leq \pi$$

$$0.95 \leq |H(e^{j\omega})| \leq 1.05, \quad 0 \leq |\omega| \leq 0.19\pi.$$

Calculate the **optimum value of ripple, attenuation and window length**.

- **This is the only paper where $\delta_p \neq \delta_s$ changes the answer**, and it is exactly the two marks for "optimum value of ripple":

$$\delta_p = 1.05 - 1 = 0.05, \quad \delta_s = 0.01$$

$$\delta = \min(\delta_p, \delta_s) = \min(0.05, 0.01) = \mathbf{0.01}$$

- **Take the smaller, never the average and never δ_p** . Kaiser's window is symmetric in its effect, so the tighter of the two requirements governs; using 0.05 gives $A = 26$ dB and a filter that fails its own stop band.

$$A = -20 \log_{10}(0.01) = \mathbf{40 \text{ dB}} \quad (\text{the "attenuation" mark})$$

$$\Delta\omega = 0.21\pi - 0.19\pi = 0.02\pi = 0.062832$$

$$M = \frac{A - 8}{2.285 \Delta\omega} = \frac{32}{0.14357} = 222.89 \Rightarrow M = 223, \quad \boxed{N = 225} \quad (\text{the "window length" mark})$$

- It never asks for $h[n]$, so stop here. $\beta = 3.3953$ and $\omega_c = 0.2\pi$ are worth one line each in case the examiner wants them.

3.3 78 Bh and 74 Ch are band pass, not low pass

78 Bh **8**, 74 Ch **9+3** · **three bands, two transitions, and you must pick one**

QUESTION AS PRINTED 2078 Bhadra · 2074 Chaitra · [8] [9+3]

Design a linear phase FIR filter using KAISER window to meet the following specifications:

$$|H(e^{jw})| \leq 0.01; \quad 0 \leq |w| \leq 0.25\pi$$

$$0.95 \leq |H(e^{jw})| \leq 1.05; \quad 0.35\pi \leq |w| \leq 0.6\pi$$

$$|H(e^{jw})| \leq 0.01; \quad 0.65\pi \leq |w| \leq \pi$$

74 Ch adds "Also determine the minimum length $(M + 1)$ of the impulse response and Kaiser window parameter β ."

- **Step that is missed #1: a band pass has TWO transitions, and the NARROWER one sets the length**. One window serves both edges, so a filter long enough for the wide transition is too short for the narrow one. Write both out:

$$\Delta\omega_1 = 0.35\pi - 0.25\pi = 0.10\pi \quad (\text{lower edge})$$

$$\Delta\omega_2 = 0.65\pi - 0.6\pi = 0.05\pi \quad (\text{upper edge})$$

$$\Delta\omega = \min(\Delta\omega_1, \Delta\omega_2) = \min(0.10\pi, 0.05\pi) = \mathbf{0.05\pi} = 0.15708$$

$$\delta = \min(\delta_p, \delta_s) = \min(0.05, 0.01) = 0.01, \quad A = 40 \text{ dB}, \quad \beta = 3.3953$$

$$M = \frac{A - 8}{2.285 \Delta\omega} = \frac{32}{2.285(0.15708)} = 89.15 \Rightarrow M = 90, \quad \boxed{N = 91}, \quad \alpha = 45$$

- **Taking the lower transition alone gives $M = 44.58$ and $N = 47$, and it is wrong** – that filter misses the 0.6π to 0.65π edge by a factor of two. It is the single easiest mark to lose in this section, because 0.25π and 0.35π are the first two numbers printed.
- **Step that is missed #2: the ideal response is the band-pass one**, not the low-pass one, and each edge again sits in the middle of its own transition:

$$\omega_1 = \frac{0.25\pi + 0.35\pi}{2} = \mathbf{0.3\pi}, \quad \omega_2 = \frac{0.6\pi + 0.65\pi}{2} = \mathbf{0.625\pi}$$

$$h_d[n] = \frac{\sin(\omega_2(n - \alpha)) - \sin(\omega_1(n - \alpha))}{\pi(n - \alpha)}, \quad h_d[\alpha] = \frac{\omega_2 - \omega_1}{\pi}$$

- The window is the same Kaiser as §3; only h_d changed.

3.4 73 Ch prints a pass band that swallows its own stop band

73 Ch 8 · no filter exists as printed

QUESTION AS PRINTED 2073 Chaitra · [8]

Design an FIR linear phase filter using Kaiser window to meet the following specifications:

$$0.98 \leq |H(e^{j\omega})| \leq 1.02, \quad \text{for } 0 \leq \omega \leq 0.9\pi$$

$$|H(e^{j\omega})| \leq 0.01, \quad \text{for } 0.14\pi \leq \omega \leq \pi$$

- **The step that is missed: read the spec before designing.** It gives $\omega_p = 0.9\pi$ and $\omega_s = 0.14\pi$, so $\omega_p > \omega_s$ and the two bands overlap. $\Delta\omega$ would be negative and M negative with it.

$$\Delta\omega = \omega_s - \omega_p = 0.14\pi - 0.9\pi = -0.76\pi < 0 \Rightarrow \text{impossible}$$

- Read 0.9π as 0.09π , a dropped zero. The transition 0.09π to 0.14π is then sensible. Note the misprint in one line, then:

$$\delta_p = 1.02 - 1 = 0.02, \quad \delta_s = 0.01, \quad \delta = \min(0.02, 0.01) = 0.01$$

$$A = 40 \text{ dB}, \quad \beta = 3.3953$$

$$\Delta\omega = 0.14\pi - 0.09\pi = 0.05\pi = 0.15708, \quad M = \frac{32}{2.285(0.15708)} = 89.15 \Rightarrow M = 90, \quad \boxed{N = 91}$$

$$\omega_c = \frac{0.09\pi + 0.14\pi}{2} = \mathbf{0.115\pi}, \quad \alpha = 45, \quad h[45] = 0.115$$

3.5 78 Ch is the only paper whose δ is not 0.01 or 0.035

78 Ch 8 · so it is the only one with a different β

QUESTION AS PRINTED 2078 Chaitra · [8]

Design a FIR linear phase filter using Kaiser window to meet the following specifications:

$$0.98 \leq |H(e^{j\omega})| \leq 1.02; \quad 0 \leq \omega \leq 0.19\pi$$

$$|H(e^{j\omega})| \leq 0.02; \quad 0.21\pi \leq \omega \leq \pi$$

- **Both ripples are 0.02 here**, so the min does not rescue you into the familiar $\beta = 3.3953$. Recompute it:

$$\delta_p = 1.02 - 1 = 0.02, \quad \delta_s = 0.02, \quad \delta = 0.02$$

$$A = -20 \log_{10}(0.02) = \mathbf{33.98 \text{ dB}}$$

$$\beta = 0.5842(33.98 - 21)^{0.4} + 0.07886(33.98 - 21) = 0.5842(12.98)^{0.4} + 0.07886(12.98) = \mathbf{2.6523}$$

$$\Delta\omega = 0.21\pi - 0.19\pi = 0.02\pi = 0.062832$$

$$M = \frac{A - 8}{2.285 \Delta\omega} = \frac{33.98 - 8}{2.285(0.062832)} = \frac{25.98}{0.14357} = 180.95 \Rightarrow M = 181, \quad N = 182 \Rightarrow \boxed{N = 183}$$

$$\omega_c = \frac{0.19\pi + 0.21\pi}{2} = \mathbf{0.2\pi}, \quad \alpha = 91$$

- **Looser ripple gives a SHORTER filter**, 183 against 75 Ash's 225 on the same transition width. That one-line comparison is often the last mark.

3.6 66 Ma, the only Kaiser paper worked end to end, with its own Bessel table

66 Ma · five parts, a 9-tap filter, and a table that gives the answer away

QUESTION AS PRINTED 2066 Magh · [parts a-e]

Design a digital low-pass FIR filter with the following specifications using Kaiser Window. Pass band frequency (ω_p) = 0.25π , stop band frequency (ω_s) = 0.65π , Pass band ripple (δ_p) = 0.035, Stop band ripple (δ_s) = 0.035.

- a) Find the order of filter (N) b) Find the cutoff frequency (ω_c)
 c) Find the value of shape parameter (β) d) Find Kaiser window ($w(n)$)
 e) Find the filter impulse response ($h(n)$)

Some modified Bessel function values are as given below.

x	0	1.3165	1.7237	1.8455	1.9271	1.93	1.9903	2
$J_0(x)$	1	1.4826	1.8926	2.0508	2.1675	2.1718	2.2642	2.2796

- **The table is mislabelled, and saying so is worth a mark.** It is headed $J_0(x)$, but the ordinary Bessel J_0 oscillates and never exceeds 1, while every value here climbs past 2. These are $I_0(x)$, the **modified** Bessel function, which is what the Kaiser window needs. Right table, wrong name.
- **And the table hands you β .** Its x column contains 1.9903, which is exactly the β this spec produces, with $I_0(\beta) = 2.2642$ ready to use.

(c) the shape parameter, and (a) the order

$$\delta = \min(\delta_p, \delta_s) = \min(0.035, 0.035) = \mathbf{0.035}$$

$$A = -20 \log_{10}(0.035) = \mathbf{29.1186 \text{ dB}}$$

$$\beta = 0.5842(29.1186 - 21)^{0.4} + 0.07886(29.1186 - 21) = \mathbf{1.9903}$$

$$\Delta\omega = 0.65\pi - 0.25\pi = 0.40\pi = 1.25664$$

$$M = \frac{A - 8}{2.285 \Delta\omega} = \frac{29.1186 - 8}{2.285(1.25664)} = \frac{21.1186}{2.87142} = 7.355 \Rightarrow M = 8, \quad \boxed{N = 9}, \quad \alpha = 4$$

(b) the cut-off

$$\omega_c = \frac{\omega_p + \omega_s}{2} = \frac{0.25\pi + 0.65\pi}{2} = \mathbf{0.45\pi} = 1.41372 \text{ rad/sample}$$

(d) and (e), from the paper's own table

$$x(n) = \beta \sqrt{1 - \left(\frac{2n}{M} - 1\right)^2} = 1.9903 \sqrt{1 - \left(\frac{2n}{8} - 1\right)^2}, \quad w[n] = \frac{I_0(x(n))}{I_0(1.9903)} = \frac{I_0(x(n))}{2.2642}$$

$$h_d[n] = \frac{\sin(0.45\pi(n-4))}{\pi(n-4)}, \quad h_d[4] = \frac{\omega_c}{\pi} = 0.45$$

n	0	1	2	3	4	5	6	7	8
$x(n)$	0	1.3165	1.7237	1.9271	1.9903	1.9271	1.7237	1.3165	0
$I_0(x)$	1	1.4826	1.8926	2.1675	2.2642	2.1675	1.8926	1.4826	1
$w[n]$	0.4417	0.6548	0.8359	0.9573	1	0.9573	0.8359	0.6548	0.4417
$h_d[n]$	-0.0468	-0.0945	0.0492	0.3144	0.4500	0.3144	0.0492	-0.0945	-0.0468
$h[n]$	-0.0207	-0.0619	0.0411	0.3010	0.4500	0.3010	0.0411	-0.0619	-0.0207

- **Only five of the paper's eight x entries are used:** 0, 1.3165, 1.7237, 1.9271, 1.9903. The other three, 1.8455, 1.93 and 2, are never reached by $x(n)$ for $M = 8$. **Do not force them into the table** to "use up" the data.
- $x(n)$ is symmetric about $n = 4$, so the window is, so $h[n]$ is: compute $n = 0$ to 4 and mirror. That halves the work and is the check.
- The centre tap is $h[4] = w[4] h_d[4] = 1 \times 0.45 = \mathbf{0.45} = \omega_c/\pi$ exactly, which is the standing check from the top of this chapter.

4 The bookwork, which is a third of the chapter

TOP 31/45 · Gibbs TOP 8/45 and Remez TOP 23/45 · 175 marks of Remez alone, and not one calculation

- **Gibbs** is asked in eight papers for 3 to 6 marks. Three sentences: truncating with a rectangular window convolves the ideal response with a kernel whose sidelobes are only 13 dB down; the result overshoots the band edge by about 9 % with ringing either side; and **lengthening the filter does not reduce the overshoot**, only narrows the ripples, so the cure is a tapered window.
- **Remez** is asked in twenty-three, for 4 to 10 marks, and almost always wants **the flowchart**. Learn the seven boxes in §5.5 of the theory: input, initial guess of $r + 2$ extremal frequencies, solve for δ , interpolate and evaluate $E(\omega)$ on a dense grid, exchange the extrema, test against the alternation theorem, output.
- The three "why is it better" marks: **it wastes no tolerance**, so it gives a **shorter filter** for the same spec; it allows **different passband and stopband ripple** through the weight $W(\omega)$; and it ignores the transition band instead of constraining it.
- **Write $\min_h \max_\omega |W(\omega)[H_d(\omega) - H(\omega)]|$ once** and the "mathematical expression" marks are collected.

Abbreviations used in these notes

Only these are ever shortened. Anything not on this list is written out in full. $\rightarrow$ reads "leads to / therefore", $+$ reads "and", $=$ reads "is". $\checkmark$ means the property holds, $\times$ that it does not.

Short	Means	Short	Means
DSP	digital signal processing	DSAP	digital signal analysis and processing
DT	discrete-time	CT	continuous-time
LTI	linear time-invariant	LSI	linear shift-invariant, the same thing
TI	time-invariant	BIBO	bounded input, bounded output
LCCDE	linear constant-coefficient difference equation	ROC	region of convergence
DTFS	discrete-time Fourier series	DTFT	discrete-time Fourier transform
CTFT	continuous-time Fourier transform	DFS	discrete Fourier series
DFT	discrete Fourier transform	IDFT	inverse DFT
FFT	fast Fourier transform	DIT	decimation in time
DIF	decimation in frequency	MAC	multiply and accumulate
FIR	finite impulse response	IIR	infinite impulse response
LPF	low pass filter	HPF	high pass filter
BPF	band pass filter	BSF	band stop filter
BLT	bilinear transformation	II	impulse invariance
A/D	analog to digital	D/A	digital to analog
S/H	sample and hold	ZOH	zero-order hold
SNR	signal to noise ratio	PYQ	past-year question
coeff	coefficient	freq	frequency
eqn	equation	no.	number
ops	operations	attr	attribute
w/	with	w/o	without
Eg	for example	Vs	compare / differentiate
+Fig	a figure is wanted	(I)	importance / need
(+)	advantage	(-)	disadvantage

Exam paper codes

Month codes

- **Ba** Baishakh **Jth** Jestha **Asa** Ashad
- **Shr** Shrawan **Bh** Bhadra **Ash** Ashwin
- **Ka** Kartik **Mng** Mangsir **Po** Poush
- **Ma** Magh **Ch** Chaitra
- **Ash and Asa are different months:** 2075 Ashwin and 2070 Ashad are separate papers.

How a year code is written

- **Bold** = **Regular** exam. Plain = Back exam.
 - “New Back (2066 & Later Batch)” counts as Back.
 - A header printing “Regular / Back” in one box counts as Regular.
- Typewriter = the EX 753 paper (DSP, BEX, IV/II).
Normal font = **CT 704** (DSAP, BEI/BCT, IV/I).
- The two axes are independent: **76 Ch** is CT 704 Regular, 76 Ash is CT 704 Back, 76 Bh is EX 753 Regular, 74 Ma is EX 753 Back.
- **69 Bh and 69 Bh are two different papers**, one in each course.

Frequency chips

- **TOP n/45** asked in **10 or more** of the 45 papers.
- **HOT n/45** asked in **4 to 9** papers.
- **PIN n/45** asked in **1 to 3** papers.
- **verified/added** written or corrected against a reference because the source notes did not cover it, or covered it wrongly.
- **5** marks the question has actually carried. Several chips mean the same question has been set at different marks in different years.
- The count is over **45 papers**: 28 CT 704 (2066 Magh to 2082 Bhadra) and 17 EX 753 (2069 Bhadra to 2081 Chaitra).
- Counts come from the Detailed PYQ document, which reproduces every question in full.